

Zhuoyun Du

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1 RESEARCH INTERESTS

My research asks whether agents can **learn to design their own communication protocols, coordination rules, and improvement mechanisms**, rather than relying on human-specified ones. I pursue this through three connected threads:

1. **Latent-Space Communication for Agentic Foundation Models:** Treating latent representations as a native medium for reasoning and coordination among agents, and as a bridge between agent design and core representation learning.
2. **Self-Improving Multi-Agent Systems:** Building agents that refine coordination and problem-solving through interaction, feedback, and textual parameter optimization, moving beyond fixed workflows.
3. **Autonomous Evaluation & Post-Training:** Exploring agent-driven evaluation, preference alignment, and data-generation loops that let models improve in open-ended settings with less dense human supervision.

2 EDUCATION

Zhejiang University, Hangzhou, China
M.Eng. in Data Science and Engineering.
State Key Lab of CAD & CG. Advisor: Prof. Wei Chen.

Sep 2024 -- Jul 2027 (Expected)
GPA: 89.5 / 100 (Overall Ranking: 4 / 50).

Jinan University, Guangzhou, China
B.Eng. in Computer Science and Technology.

Sep 2020 -- Jul 2024
GPA: 90 / 100; (Rank: 3 / 99).

3 SELECTED PUBLICATIONS

† indicates co-first / equal contribution.

First-author / co-first-author

1. **Zhuoyun Du**[†], Runze Wang[†], Huiyu Bai, Zouying Cao, Xiaoyong Zhu, Bo Zheng, Wei Chen and Haochao Ying. Enabling Agents to Communicate Entirely in Latent Space [**Interlat**]. *ACL 2026*. [paper] [Code]
2. **Zhuoyun Du**[†], Lujie Zheng[†], Renjun Hu, Yuyang Xu, Xiawei Li, Ying Sun, Wei Chen, Jian Wu, Haolei Cai, and Haochao Ying. LLMs can simulate standardized patients via agent coevolution [**EvoPatient**]. *ACL 2025*. [paper] [Code]
3. **Zhuoyun Du**[†], Chen Qian[†], Wei Liu, Zihao Xie, Yifei Wang, Yufan Dang, Weize Chen, and Cheng Yang. Multi-agent software development through cross-team collaboration [**Croto**]. *Findings of ACL 2025*. [paper] [Code]
4. Huiyu Bai[†], Runze Wang[†], **Zhuoyun Du**[†], Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng and Xuejiao Zhao. Online-PVLM: Advancing Personalized VLMs with Online Concept Learning [**Online-PVLM**]. *Submitted to MM 2026*. [arXiv]

Co-authored

1. Shan He, Runze Wang, **Zhuoyun Du**, Huiyu Bai, Zouying Cao, Yu Cheng, Bo Zheng. Learning to Evolve: A Self-Improving Framework for Multi-Agent Systems via Textual Parameter Graph Optimization [**Learning to Evolve**]. *Findings of ACL 2026*. [paper]
2. Yuewen Liu, Peng Xu, **Zhuoyun Du**, Muxi Diao, Anyi Zhang, Yang Li, Yutong Zhang. MemCoRL: Alternating Co-Optimization of Memory Retrieval and Utilization via Collaborative Reinforcement Learning [**MemCoRL**]. *ACL 2026*.
3. Chen Qian[†], Zihao Xie[†], Yifei Wang[†], Wei Liu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, Zhiyuan Liu, and Maosong Sun. Scaling large-language-model-based multi-agent collaboration [**MACNet**]. *ICLR 2025*. [paper]
4. Wei Liu[†], Chenxi Wang[†], Yifei Wang, Zihao Xie, Rennai Qiu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, and Chen Qian. Autonomous agents for collaborative task under information asymmetry [**iAgent**]. *NeurIPS 2024*. [paper]

4 RESEARCH & INDUSTRY EXPERIENCE

ByteDance, Data Department
Talent Program Intern.

Mar 2026 -- Present

- Built the team's **first** data-flywheel rollout system for multi-turn, task-oriented dialogue, supporting downstream **multi-turn dialogue RL** and **agentic RL**.
- Designed a **self-play simulation and evaluation environment** with high-fidelity, persona-conditioned user-simulation agents for rollout generation and evaluation signals.
- Conducted research on **LLM post-training and preference alignment** (SFT, RLHF/DPO) and autonomous agent-driven evaluation of dialogue policies.

Alibaba, Future Living Lab / Token Foundry with Tongyi
Research Intern.

Jan 2025 -- Mar 2026

- Worked in a foundation-model ecosystem whose broader initiatives include HappyHorse (#1 on Artificial Analysis Video Arena) and HappyOyster (real-time interactive world model), focusing on latent-space communication for multi-agent reasoning.
- Proposed **Interlat**, replacing natural-language message passing with temporally aligned hidden-state exchange; designed token-latent curriculum and learned compression, achieving up to 24× speedup and gaining hands-on 256+ GPU distributed training experience.

- Co-developed **Online-PVLM Submitted to MM 2026**; co-authored **MemCoRL ACL 2026** and **Learning to Evolve Findings of ACL 2026**.

Zhejiang University, State Key Lab of CAD & CG

Sep 2024 -- Present

M.Eng. Research, advised by Prof. Wei Chen.

- Led **EvoPatient ACL 2025**, an open-source doctor-patient coevolution system for virtual standardized patients, piloted in clinical teaching and OSCE-style training at hospitals including the Second Affiliated Hospital of Zhejiang University School of Medicine.
- Drove research formulation, hospital-side requirement gathering, coevolution-method design, implementation, prompt engineering, experiments, medical-case curation/de-identification, expert evaluation, and paper writing.
- Improved patient-answer *Ability* score from 0.763 (Few-shot baseline) to 0.860 after 200 training cases. [\[paper\]](#)

Tsinghua University, THUNLP

Dec 2023 -- Aug 2024

Research Assistant.

- Conducted research within the **OpenBMB/ChatDev** ecosystem (**33k+ GitHub stars**) under Prof. **Zhiyuan Liu**, studying role organization, information exchange, and coordination among LLM agents.
- Led **Croto Findings of ACL 2025** as a ChatDev-branch project, designing cross-team orchestration with parallel teams, hierarchy partitioning, and greedy aggregation.
- Contributed to **MACNet ICLR 2025** and **iAgent NeurIPS 2024**, covering scalable collaboration and information-asymmetry settings.

SKILLS, HONORS, SERVICE & BACKGROUND

- **LLM & Machine Learning:** Multi-agent system design; distributed training on **256+ GPUs** (e.g., A100, A800, H100, H20); SFT, LoRA, RLHF/DPO; DeepSpeed, FlashAttention.
- **Programming Languages & Frameworks:** Python, C/C++, Bash, Git, $\LaTeX$, PyTorch, HuggingFace ecosystem (e.g., Transformers, Datasets).
- **Languages:** English (IELTS: 7.5); Mandarin (Native).
- **Honors:**
 - 2025 **National Scholarship**
 - Outstanding Graduate Student Scholarship
 - 3 Undergraduate **Scholarships**
 - **First-Class Scholarship for Outstanding Graduates**
- **Service:** Program Committee, NeurIPS 2025 and ICLR 2025.
- **Interests & Background:** Experienced in collaborative research and communication; former **National Champion in Amateur Épée Fencing**; Grade 7 Piano Certificate; interests include snowboarding, photography, and arts and culture.